Chulalongkorn University · Department of Chemical Engineering

Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

Equipment list, datasheets and wall-thickness checks

NDFH-V1-ME-EQL-001 · Rev 0 · Issued for teaching

Rev	Date	Description	Originator	Checker	Approver
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Title block	
Client	Chulalongkorn University · Department of Chemical Engineering
Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator
Document no.	NDFH-V1-ME-EQL-001
Title	Equipment list, datasheets and wall-thickness checks
Drawings	PFD-100 rev 1; P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev E (EPC alternative)
Scope	Equipment list, process datasheets of H-100, E-100 and P-100A/B, design conditions, wall thickness and a general arrangement sketch.

All numbers in this document come from the Natural Draft Fired Heater Ver 1 simulator model (design case and model runs) or are stated as assumed. It is a teaching document, not a design for construction.

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1 · Equipment list

Tag	Name	Type	Size	Duty	Design P	Design T	Materials	DS
H-100	Fired heater, natural draft, vertical cylindrical	API 560, 1 burner, 1 pass	Radiant 42 × 3 in sch 40 × 4.5 m on Ø2.38 m; convection 28 tubes (8 shield); stack 7.3 m × Ø0.64 m	2312 kW absorbed / 2835 kW fired	Coil 1700 kPa(g)	Fluid 380 °C; tube metal 425 °C	Coil SA-106 B; casing CS; refractory castable + ceramic fibre	ME-DS-101
B-101	Oil burner (in H-100)	Natural-draft oil burner, steam- or mechanical-atomised (vendor)	Max release 3.48 MW (110 % of the maximum case)	2835 kW fired at design	Gun 600 kPa(g) at design, 1400 max	—	Vendor	ME-DS-102
E-100	Feed preheater, shell and tube	TEMA AES (A6), feed in tubes, heating fluid in shell	82 m² (71 m² + 15 %), ~225 tubes ¾ in × 6.1 m	1904 kW	Tube 1700 / shell 1000 kPa(g)	Tube 230 / shell 260 °C	CS shell, CS tubes	ME-DS-103
P-100A/B	Heater feed pumps (1 running + 1 spare)	Centrifugal, API 610 OH2	Rated 27 m³/h × 115 m; shut-off 172 m	Motor 15 kW each	Casing 1700 kPa(g)	Design 80 °C	CS casing, 12 % Cr impeller	ME-DS-104
F-100	Combustion air intake with screen	Louvred intake, bird and debris screen, on the windbox	3867 kg/h at design; 4312 kg/h maximum case	—	Atmospheric	Ambient	CS galvanised	—
HV-400	Stack damper	Single-blade butterfly, fail open, with minimum-open stop	Ø0.64 m stack	—	Atmospheric	500 °C	CS with SS shaft	—

2 · Process datasheets

ME-DS-101 · H-100 fired heater

Item	Value
Service	Heat feed 180 → 350 °C for V-200
Type	Vertical cylindrical, natural draft, 1 burner
Absorbed duty, design / maximum	2312 / 2550 kW
Fired duty (LHV), design	2835 kW
Efficiency (LHV), design	81.6 %
Radiant / convection duty	1590 / 722 kW
Radiant tubes	42 × 3 in sch 40 (OD 88.9 mm), effective length 4.53 m, pitch 2 × OD
Tube circle diameter	2.38 m (H/D 1.91)
Average radiant flux	30 kW/m² (design basis 30)
Tube skin, average / peak (calculated)	383 / 409 °C
Convection tubes	28 (8 shield), 7 rows
Mass velocity in the coil	1141 kg/m² s (1.43 m/s)
Draft at the floor / required	6.7 / 5 mmH_2O
Stack	7.26 m × Ø0.64 m
Excess air / O_2 (firebox)	11 % / 2 %
Design pressure / temperature (coil)	1700 kPa(g) / 380 °C fluid, 425 °C metal

ME-DS-103 · E-100 feed preheater

Item	Value
Duty	1904 kW
Tube side (feed)	19584 kg/h, 40 → 180 °C
Shell side (heating fluid, A4)	37250 kg/h, 230 → 150 °C
LMTD (counter-current)	76.1 K
U (A5)	350 W/m² K
Area required / installed	71.5 / 82.2 m²
Model effectiveness at HV-063 60 %	73.7 % (TI-010 180 °C)
Design pressure tube / shell	1700 / 1000 kPa(g)
Design temperature tube / shell	230 / 260 °C

ME-DS-104 · P-100A/B feed pumps

Item	Value
Liquid	Heater feed, ρ 800 kg/m³ at 40 °C
Flow normal / rated	24.5 / 27 m³/h
Suction pressure (PI-001)	50 kPa(g)
Discharge at rated (pump curve)	950 kPa(g)
Differential head at rated	114.7 m
Shut-off pressure / head	1400 kPa(g) / 172 m
Hydraulic / absorbed power at rated	6.75 / 11.3 kW (η 60 %, A3)
Motor	15 kW, 400 V
NPSH available / required	18.6 / 3 m

3 · Design pressure and temperature

Design pressure = max(1.1 × maximum operating, maximum operating + 170 kPa), rounded up to 100 kPa. Design metal temperature of a fired tube follows API 530: highest calculated or measured tube-metal temperature + 15 °C.

Equipment	Design pressure	Design temperature
H-100 coil	1700 kPa(g) (maximum operating 1450)	380 °C fluid, 425 °C metal
E-100 tube side	1700 kPa(g)	230 °C
E-100 shell side (A4)	1000 kPa(g)	260 °C
P-100A/B casing	1700 kPa(g)	80 °C
Fuel oil train	2200 kPa(g)	150 °C

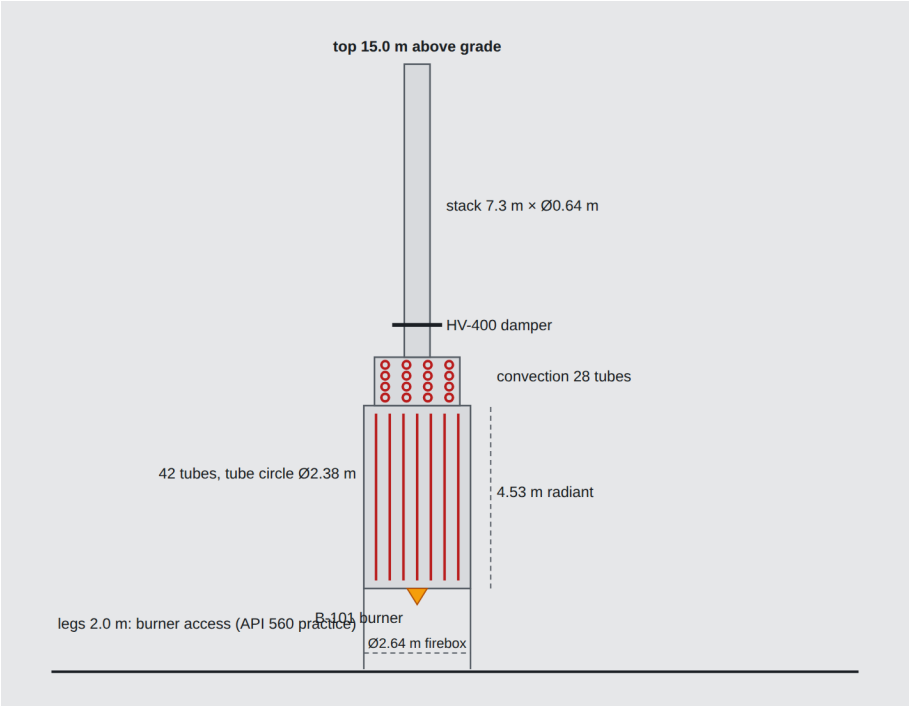
4 · Wall thickness

Item	Formula	P MPa	D mm	Material at T	S MPa	E	CA	t req. mm	Supplied mm	Result
Radiant tube, 3 in sch 40	API 530 elastic, $t = P D_o/(2S + P) + CA$	1.70	88.9	SA-106 B at 425 °C	75	1	3	3.99	5.49	✓
E-100 shell (ID 590 mm)	ASME VIII-1 UG-27, $t = P R/(S E − 0.6P) + CA$	1.00	590.0	SA-516-70 at 260 °C	138	0.85	3	5.53	10.00	✓
Outlet line 3"-P-105	ASME B31.3 304.1.2, $t = P D/(2(S E + P Y)) + CA$	1.70	88.9	SA-106 B at 380 °C	102	1	3	3.73	4.80	✓

Illustrative allowable stresses (A9), MPa

Material	343 °C	371 °C	399 °C	427 °C	454 °C	482 °C	510 °C
SA-106 B (CS pipe)	118	108	90	74	59	—	—
SA-516-70 (CS plate)	138	127	103	83	64	—	—
SA-335 P11 (1¼Cr-½Mo)	113	111	108	104	98	82	—
SA-213 TP304	108	—	104	—	100	—	95

5 · General arrangement sketch



From the Design tab sizing; a teaching sketch, not a vendor drawing.

Assumptions used in this document

No.	Assumed	Value	Used in
A1	Feed viscosity	1.0 mPa s at 40 °C, 0.5 mPa s at 180–350 °C (Design tab value)	Line sizing
A2	Feed vapour pressure at 40 °C	5 kPa(a)	NPSH available
A3	Pump efficiency P-100A/B	60 % at rated flow	Absorbed power, motor size
A4	Heating fluid from Unit A	hot oil, ρ 850 kg/m³, c_p 2.3 kJ/kg K, return 150 °C, operating 600 kPa(g), design 1000 kPa(g) / 260 °C	E-100 rating, tube-rupture case
A5	E-100 overall U	350 W/m² K (liquid–liquid, clean), 15 % area margin	E-100 area
A6	E-100 geometry	TEMA AES, ¾ in 14 BWG tubes (ID 14.8 mm), shell OD 0.61 m × 6.1 m	Wetted area, rupture area
A7	Fuel oil	ρ 900 kg/m³, μ 10 mPa s, delivered heated to 120 °C, flash point 66 °C	Fuel line, area classification
A8	Feed fluid	flash point 45 °C, auto-ignition 250 °C, latent heat 300 kJ/kg, vapour M 150, k 1.05 at relief	Relief and area classification
A9	Allowable stresses	illustrative ASME II-D values (see the stress table); confirm from the code edition in force	Wall thickness
A10	Electrical	400 V, 3-phase, 50 Hz; motor η 92 %, pf 0.86; starting current 6 × FLC	Load list, cable sizing
A11	Lead times	typical market lead times in weeks, for teaching	Procurement schedule
A12	Corrosion allowance	3 mm carbon steel coil and piping, 3 mm vessels	Wall thickness

6 · Checks

Check	This document	Compared with	Result
Radiant tubes = Design tab sizing	Radiant 42 × 3 in sch 40 × 4.5 m on Ø2.38 m	42 tubes × 4.5 m	✓
Coil design pressure ≥ pump shut-off + suction	1700 kPa(g)	1450 kPa(g)	✓
Coil design fluid temperature ≥ TSHH-025 setting	380 °C	375 °C (trip setting)	✓
Tube design metal temperature ≥ highest TMT + 15 °C	425 °C	409 °C + 15	✓
Tube skin alarm TI-024 below the design metal temperature	430 °C	425 °C	✗
E-100 tube design temperature ≥ hottest feed possible (HV-063 100 %)	230 °C	220.5 °C (model relation)	✓
E-100 tube design pressure = PSV-101 set	1700 kPa(g)	1700 kPa(g)	✓
Wall: Radiant tube, 3 in sch 40	3.99 mm required	5.49 mm supplied	✓
Wall: E-100 shell (ID 590 mm)	5.53 mm required	10.00 mm supplied	✓
Wall: Outlet line 3"-P-105	3.73 mm required	4.80 mm supplied	✓
Burner rating covers the maximum case + 10 %	3.48 MW rating	3161 kW fired at maximum × 1.1	✓